

Q1 JEE Main 2020 - 7 Jan (Morning)

If the sum of the coefficients of all even powers of x in the product $(1 + x + x^2 + \dots + x^{2n})(1 - x + x^2 - x^3 + \dots + x^{2n})$ is 61, then n is equal to

Q2 JEE Main 2020 - 7 Jan (Evening)

The number of ordered pairs (r, k) for which $6 \cdot {}^{35}C_r = (k^2 - 3) \cdot {}^{36}C_{r+1}$, where k is an integer, is :

- (A) 6
- (B) 2
- (C) 3
- (D) 4

Q3 JEE Main 2020 - 7 Jan (Evening)

The coefficient of x^7 in the expression $(1 + x)^{10} + x(1 + x)^9 + x^2(1 + x)^8 + \dots + x^{10}$ is :

- (A) 330
- (B) 210
- (C) 420
- (D) 260

Q4 JEE Main 2020 - 8 Jan (Morning)

If a , b and c are the greatest values of ${}^{19}C_p$, ${}^{20}C_q$ and ${}^{21}C_r$ respectively, then :

- (A) $\frac{a}{11} = \frac{b}{22} = \frac{c}{42}$
- (B) $\frac{a}{22} = \frac{b}{11} = \frac{c}{42}$
- (C) $\frac{a}{22} = \frac{b}{42} = \frac{c}{11}$
- (D) $\frac{a}{21} = \frac{b}{11} = \frac{c}{22}$

Q5 JEE Main 2020 - 8 Jan (Evening)

If α and β be the coefficient of x^4 and x^2 respectively in the expansion of

$\left(x + \sqrt{x^2 - 1}\right)^6 + \left(x - \sqrt{x^2 - 1}\right)^6$, then :

- (A) $\alpha + \beta = 110$
- (B) $\alpha + \beta = -150$
- (C) $\alpha - \beta = -132$

(D) $\alpha - \beta = -60$

Q6 JEE Main 2020 - 9 Jan (Morning)

Find the coefficient of x^4 in $(1 + x + x^2)^{10}$

Q7 JEE Main 2020 - 9 Jan (Evening)

In the expansion of $\left(\frac{x}{\cos \theta} + \frac{1}{x \sin \theta}\right)^{16}$, if l_1 is the least value of the term independent of x when $\frac{\pi}{8} \leq \theta \leq \frac{\pi}{4}$ and l_2 is the least value of the term independent of x when $\frac{\pi}{16} \leq \theta \leq \frac{\pi}{8}$, then the ratio $l_2 : l_1$ is equal to:

- (A) 1 : 8
- (B) 1 : 16
- (C) 8 : 1
- (D) 16 : 1

Q8 JEE Main 2020 - 9 Jan (Evening)

If $C_r = {}^{25}C_r$ and $C_0 + 5 \cdot C_1 + 9 \cdot C_2 + \dots + (101) \cdot C_{25} = 2^{25} \cdot k$, then k is equal to

Q9 JEE Main 2020 - 2 Sep (Morning)

Let $\alpha > 0, \beta > 0$ be such that $\alpha^3 + \beta^2 = 4$. If the maximum value of the term independent of x in the binomial expansion of $\left(\alpha x^{\frac{1}{9}} + \beta x^{-\frac{1}{6}}\right)^{10}$ is $10k$, then k is equal to :

- (A) 84
- (B) 176
- (C) 336
- (D) 352

Q10 JEE Main 2020 - 2 Sep (Evening)

For a positive integer n , $\left(1 + \frac{1}{x}\right)^n$ is expanded in increasing powers of x . If three consecutive coefficients in this expansion are in the ratio, $2 : 5 : 12$, then n is equal to

Q11 JEE Main 2020 - 3 Sep (Morning)

The value of $(2^1 P_0 - 3^2 P_1 + 4^3 P_2 - \dots \text{up to } 51^{\text{th}} \text{ term}) + (1! - 2! + 3! - \dots \text{up to } 51^{\text{th}} \text{ term})$ is equal to :

- (A) $1 + (52)!$
(B) $1 + (51)!$
(C) $1 + 51(51)!$
(D) $1 + (51)!$

Q12 JEE Main 2020 - 3 Sep (Morning)

If the number of integral terms in the expansion of $\left(3^{\frac{1}{2}} + 5^{\frac{1}{8}}\right)^n$ is exactly 33, then the least value of n is :

- (A) 264
(B) 128
(C) 256
(D) 248

Q13 JEE Main 2020 - 3 Sep (Evening)

If the term independent of x in the expansion of $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ is k , then $18k$ is equal to :

- (A) 9
(B) 11
(C) 5
(D) 7

Q14 JEE Main 2020 - 4 Sep (Morning)

The value of $\sum_{r=0}^{20} {}^{50}C_r - {}^{30}C_6$ is equal to:

- (A) $(1) {}^{51}C_7 - {}^{30}C_7$
(B) $(2) {}^{50}C_7 - {}^{30}C_7$
(C) $(3) {}^{51}C_7 + {}^{30}C_7$
(D) $(4) {}^{50}C_6 - {}^{30}C_6$

Q15 JEE Main 2020 - 4 Sep (Morning)

Let $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$. Then $\frac{a_7}{a_{13}}$ is equal to

Q16 JEE Main 2020 - 4 Sep (Evening)

If for some positive integer n , the coefficients of three consecutive terms in the binomial expansion of $(1+x)^{n+5}$ are in the ratio 5 : 10 : 14, then the largest coefficient in this expansion is :

- (A) 252
- (B) 462
- (C) 792
- (D) 330

Q17 JEE Main 2020 - 5 Sep (Morning)

The natural number m , for which the coefficient of x in the binomial expansion of $\left(x^m + \frac{1}{x^2}\right)^{22}$ is 1540, is

Q18 JEE Main 2020 - 5 Sep (Evening)

The coefficient of x^4 in the expansion of $(1+x+x^2+x^3)^6$ in powers of x , is

Q19 JEE Main 2020 - 6 Sep (Morning)

If $\{p\}$ denotes the fractional part of the number p , then $\left\{\frac{3^{200}}{8}\right\}$, is equal to:

- (A) $\frac{5}{8}$
- (B) $\frac{1}{8}$
- (C) $\frac{7}{8}$
- (D) $\frac{3}{8}$

Q20 JEE Main 2020 - 6 Sep (Evening)

If the constant term in the binomial expansion of $\left(\sqrt{x} - \frac{k}{x^2}\right)^{10}$ is 405, then $|k|$ equals :

- (A) 2
- (B) 3
- (C) 9
- (D) 1

Answer Key

Q1 (30)	Q2 (D)	Q3 (A)	Q4 (A)
Q5 (C)	Q6 (615)	Q7 (D)	Q8 (51)
Q9 (C)	Q10 (118)	Q11 (B)	Q12 (C)
Q13 (D)	Q14 (A)	Q15 (8)	Q16 (B)
Q17 (13)	Q18 (120)	Q19 (B)	Q20 (B)

Q1

$$\text{Let } (1 - x + x^2 \dots) (1 + x + x^2 \dots) = a_0 + a_1 x + a_2 x^2 + \dots$$

$$\text{put } x = 1$$

$$1(2n+1) = a_0 + a_1 + a_2 + \dots + a_{2n} \dots (i)$$

$$(2n+1) \times 1 = a_0 - a_1 + a_2 - \dots + a_{2n} \dots (ii)$$

$$\text{Form (i) + (ii)}$$

$$4n+2 = 2(a_0 + a_2 + \dots)$$

$$= 2 \times 61$$

$$\Rightarrow 2n+1 = 61 \Rightarrow n = 30$$

Q2

$$\frac{36}{r+1} \times \frac{35}{C_r} (k^2 - 3) = \frac{35}{C_r}$$

$$k^2 - 3 = \frac{r+1}{6} \Rightarrow k^2 = 3 + \frac{r+1}{6}$$

$$r \text{ can be } 5, 35$$

$$\text{for } r = 5, k = \pm 2$$

$$r = 35, k = \pm 3$$

$$\text{Hence number of order pair} = 4$$

Q3

$$\frac{(1+x^{10}) \left[1 - \left(\frac{x}{1+x} \right)^{11} \right]}{\left(1 - \frac{x}{1+x} \right)}$$

$$\frac{(1+x)^{10} [(1+x)^{11} - x^{11}]}{(1+x)^{11} \times \frac{1}{(1+x)}}$$

$$= (1+x)^{11} - x^{11}$$

$$\text{Coefficient of } x^7 \text{ is } {}^{11}C_7 = {}^{11}C_4 = 330$$

Q4

$$\text{We know } {}^nC_r \text{ is max at middle term}$$

$$a = {}^{19}C_p = {}^{19}C_{10} = {}^{19}C_9$$

$$b = {}^{20}C_q = {}^{20}C_{10}$$

$$c = {}^{21}C_6 = {}^{21}C_{10} = {}^{21}C_{11}$$

$$\frac{a}{{}^{19}C_9} = \frac{b}{{}^{20}C_{10}} = \frac{c}{{}^{21}C_{11}} = \frac{20}{11} \cdot \frac{19}{10} \cdot \frac{19}{10} \cdot {}^{19}C_9$$

$$\frac{a}{11} = \frac{b}{22} = \frac{c}{42/11}$$

$$\frac{a}{11} = \frac{b}{22} = \frac{c}{42}$$

Q5

$$2 \left[{}^6C_0 x^6 + {}^6C_2 x^4 (x^2 - 1) + {}^6C_4 x^2 (x^2 - 1)^2 + {}^6C_6 (x^2 - 1)^3 \right]$$

$$= 2 \left[x^6 + 15(x^6 - x^4) + 15x^2(x^4 - 2x^2 + 1) + (-1 + 3x^2 - 3x^4 + x^6) \right]$$

$$= 2(32x^6 - 48x^4 + 18x^2 - 1)$$

$$\alpha = -96 \text{ and } \beta = 36$$

$$\therefore \alpha - \beta = -132$$

Q6

General term $\frac{10!}{\alpha!\beta!\gamma!} x^{\beta+2\gamma}$

for coefficient of $x^4 \Rightarrow \beta + 2\gamma = 4$

$\gamma = 0, \beta = 4, \alpha = 6 \Rightarrow \frac{10!}{6!4!0!} = 210$

$\gamma = 1, \beta = 2, \alpha = 7 \Rightarrow \frac{10!}{7!2!1!} = 360$

$\gamma = 2, \beta = 0, \alpha = 8 \Rightarrow \frac{10!}{8!0!2!} = 45$

Total = 615

Q7

$$T_{r+1} = {}^{16}C_r \left(\frac{x}{\cos \theta} \right)^{16-r} \left(\frac{1}{x \sin \theta} \right)^r$$

for $r = 8$ term is free from x

$$T_9 = {}^{16}C_8 \frac{1}{\sin^8 \theta \cos^8 \theta}$$

$$T_9 = {}^{16}C_8 \frac{2^8}{(\sin 2\theta)^8}$$

in $\theta \in \left[\frac{\pi}{8}, \frac{\pi}{4} \right], L_1 = {}^{16}C_8 2^8$

$\therefore$ {Min value of L_1 at $\theta = \pi/4$ }

in $\theta \in \left[\frac{\pi}{16}, \frac{\pi}{8} \right], L_2 = {}^{16}C_8 \frac{2^8}{\left(\frac{1}{\sqrt{2}} \right)^8} = {}^{16}C_8 \cdot 2^8 \cdot 2^4$

{ $\therefore$ min value of L_2 at $\theta = \pi/8$ }

$$\frac{L_2}{L_1} = \frac{{}^{16}C_8 \cdot 2^8 2^4}{{}^{16}C_8 \cdot 2^8} = 16$$

Q8

$$\begin{aligned} \sum_{r=0}^{25} (4r+1) {}^{25}C_r &= 4 \sum_{r=0}^{25} r \cdot {}^{25}C_r + \sum_{r=0}^{25} {}^{25}C_r \\ &= 4 \sum_{r=0}^{25} r \times \frac{25}{r} {}^{24}C_{r-1} + 2^{25} = 100 \sum_{r=0}^{25} {}^{24}C_{r-1} + 2^{25} \\ &= 100 \cdot 2^{24} + 2^{25} = 2^{55} (50 + 1) = 51 \cdot 2^{25} \end{aligned}$$

So $k = 51$

Q9

General term of $(\alpha x^{1/9} + \beta x^{-1/6})^{10} = {}^{10}C_r (\alpha x^{1/9})^{10-r} (\beta x^{-1/6})^r$

for term independent of x $r = 4$

$\therefore$ Term independent of $x = {}^{10}C_4 \alpha^6 \beta^4$

Also $\alpha^3 + \beta^2 = 4$

By AM-GM inequality

$$\frac{\alpha^3 + \beta^2}{2} \geq (\alpha^3 \beta^2)^{1/2}$$

$$\Rightarrow (2)^2 \geq \alpha^3 \beta^2$$

$$\Rightarrow \alpha^6 \beta^4 \leq 16$$

$$\therefore 10k = 10C_4 \cdot 16$$

$$\Rightarrow k = 336$$

Q10

$${}^nC_{r-1} : {}^nC_r : {}^nC_{r+1} = 2 : 5 : 12$$

$$\Rightarrow \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{5}{2} \text{ and } \frac{{}^nC_{r+1}}{{}^nC_r} = \frac{12}{5}$$

$$\Rightarrow \frac{n-r+1}{r} = \frac{5}{2} \text{ and } \frac{n-r}{r+1} = \frac{12}{5}$$

$$\Rightarrow 2n - 7r + 2 = 0 \text{ and } 5n - 17r - 12 = 0$$

Solving ; $x = 118, r = 34$

Q11

$$\therefore (r+1) \cdot {}^r P_{r-1} = (r+1) \cdot \frac{r!}{1} = r! + 1$$

$$\text{So } (2 \cdot {}^1 P_0 - 3 \cdot {}^2 P_1 + \dots \dots \dots 51 \text{ terms}) +$$

$$(11 - |2 + |3 - \dots \dots \dots \text{ upto 51 terms })$$

$$= [|2 - |3 + |4 - \dots \dots \dots + |52] + [|1 - |2 + |3 - \dots \dots \dots + |51]$$

$$= |52 + |1 = 52 + 1$$

Q12

$$\left(3^{\frac{1}{2}} + 5^{\frac{1}{8}}\right)^n$$

$$\text{Let } T_{r+1} = {}^n C_r (3)^{\frac{n-r}{2}} (5)^{\frac{r}{8}}$$

$$\text{So } r \text{ must be } 0, 8, 16, 24, \dots$$

$$\text{Now } n = t_{33} = 0 + 32 \times 8 = 256$$

$$\Rightarrow n = 256$$

Q13

$$\text{General term} = T_r + 1 = {}^9 C_r \left(\frac{3}{2}x^2\right)^{9-r} \left(-\frac{1}{3x}\right)^r$$

$$= {}^9 C_r (-1)^r \cdot \frac{3^{9-2r}}{2^{9-r}} x^{18-3r}$$

If term is independent of x then $r = 6$.

$$\therefore k = {}^9 C_6 \cdot \frac{3^{-3}}{2^3} = \frac{7}{18}$$

$$\therefore 18k = 7$$

Q14

$$\text{Here } \sum_{r=0}^{20} {}^{50-r} C_6 = {}^{50} C_6 + {}^{49} C_6 + {}^{48} C_6 + {}^{47} C_6 + \dots + {}^{32} C_6 + {}^{31} C_6 + {}^{30} C_6$$

$$\Rightarrow ({}^{30} C_7 + {}^{30} C_6) + {}^{31} C_6 + {}^{32} C_6 + \dots + {}^{49} C_6 + {}^{50} C_6$$

$$- {}^{30} C_7$$

$$\Rightarrow ({}^{31} C_7 + {}^{31} C_6) + {}^{32} C_6 + \dots + {}^{49} C_6 + {}^{50} C_6 - {}^{30} C_7$$

$$= ({}^{32} C_7 + {}^{32} C_6) + \dots + {}^{49} C_6 + {}^{50} C_6 - {}^{30} C_7$$

$$= {}^{51} C_7 - {}^{30} C_7$$

Q15

$$(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$$

$$\text{General term} = \frac{10!}{r_1! r_2! r_3!} (2x^2)^{r_1} (3x)^{r_2} (4)^{r_3}$$

As $a_7 = \text{coeff of } x^7$

$$2r_1 + r_2 = 7 \text{ and } r_1 + r_2 + r_3 = 10$$

Possibilities are

$$\begin{array}{|c|c|c|} \hline (r_1) & (r_2) & (r_3) \\ \hline 0 & 7 & 3 \\ \hline 1 & 5 & 4 \\ \hline 2 & 3 & 5 \\ \hline 3 & 1 & 6 \\ \hline \end{array}$$

$$a_7 = \frac{10! 3^7 4^3}{7! 3!} + \frac{10! (2)(3)^5 (4)^4}{5! 4!}$$

$$+ \frac{10! (2)^2 (3)^3 (4)^5}{2! 3! 5!} + \frac{10! (2)^3 (3)(4)^6}{3! 6!}$$

$$a_{13} = \text{Coeff of } x^{13}$$

$$2r_1 + r_2 = 13 \text{ and } r_1 + r_2 + r_3 = 10$$

Possibilities are

$$\begin{array}{|c|c|c|} \hline (r_1) & (r_2) & (r_3) \\ \hline 3 & 7 & 0 \\ \hline 4 & 5 & 1 \\ \hline 5 & 3 & 2 \\ \hline 6 & 1 & 3 \\ \hline \end{array}$$

$$a_{13} = \frac{10!(2^3)(3^7)}{3!7!} + \frac{10!(2^4)(3^5)(4)}{4!5!} \\ + \frac{10!(2^5)(3^3)(4^2)}{5!3!2!} + \frac{10!(2^6)(3)(4^3)}{6!1!3!}$$

Clearly $\frac{a_7}{a_{13}} = 2^3 = 8$

Q16

Consider the three consecutive coefficients as

$${}^{n+5}C_r, {}^{n+5}C_{r+1}, {}^{n+5}C_{r+2}$$

$$\therefore \frac{n+5C_r}{n+5C_{r+1}} = \frac{1}{2}$$

$$\Rightarrow \frac{r+1}{n+5-r} = \frac{1}{2} \Rightarrow 3r = n+3$$

$$\text{and } \frac{n+5}{n+5} C_{r+2} = \frac{5}{7}$$

$$\Rightarrow \frac{r+2}{n+4-r} = \frac{5}{7} \Rightarrow 12r = 5n + 6 \quad \dots (ii)$$

From (i) and (ii) $n = 6$

Largest coefficient in the expansion is ${}^{11}C_6 = 462$

Q17

$$T_{r+1} = {}^{22}C_r \cdot (x^m)^{22-r} \cdot \left(\frac{1}{x^2}\right)^r$$

$$T_{r+1} = {}^{22}C_r \cdot x^{22m-mr-2r}$$

$$\therefore 22m - mr - 2r = 1 \text{ and } 22c_r = 1540$$

$$\therefore 22c_3 = 1540 \Rightarrow r = 3 \text{ or } 19$$

Now, $f_{orr} = 3$; $22m - 3m - 6 = 1$

$$\Rightarrow 19m = 7 \Rightarrow m = \frac{7}{19} \quad (\text{not acceptable})$$

$$\text{for } r = 19; 22m - 19m = 39$$

$$\Rightarrow m = 13$$

Q18

$$(1 + x + x^2 + x^3)^6 = \left(\frac{1-x^4}{1-x}\right)^6$$

Coefficient of x^4 in $\left(\frac{1-x^4}{1-x}\right)^6 = \text{coefficient of } x^4 \text{ in } (1-6x^4)(1-x)^{-6}$

= coefficient of x^4 in

$$(1 - 6x^4) [1 + {}^6C_1x + {}^7C_2x^2 + \dots]$$

$$= {}^9C_4 - 6.1 = 126 - 6 = 120$$

Q19

$$\frac{3^{200}}{8} = \frac{1}{8} (9^{100})$$

$$= \frac{1}{8}(1+8)^{100}$$

$$= \frac{1}{8} + \text{Integer}$$

$$\therefore \left\{ \frac{3^{200}}{8} \right\} = \left\{ \frac{1}{8} + \text{integer} \right\}$$

$$= \frac{1}{8}$$

Q20

$$\text{General term} = T_{r+1} = {}^{10}C_r (\sqrt{x})^{10-r} \cdot \left(-\frac{k}{x^2}\right)^r$$

$$= {}^{10}C_r (-k)^r \cdot x^{\frac{10-r}{2} - 2r}$$

$$= {}^{10}C_r(-k)^r \cdot x^{\frac{10-5r}{2}}$$

If it is constant term then $x = 2$

$$\therefore {}^{10}C_2(-k)^2 = 405$$

$$\Rightarrow k^2 = \frac{405 \times 2}{10 \times 9} = \frac{81}{9} = 9$$

$|k| = 3$

